Annotated Bibliography

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Liang et al. [2025]
Martin Liang, Joel Ng, Fred Huang, Louis Yong, Pi Rong Koh, Hamish Starling, and Luke Moran. First order robotics team description paper small size league of robocup 2025. RoboCup Small Size League, 2025

Aim: This paper details the redesign of the "First Order" robot to address previous mechanical and software failures in RoboCup SSL Division B. The solution includes a compact four-wheel omnidirectional drive and spring-damped dribbler, paired with a modular Python software stack using a thread-safe queue system and centralized data repository.

Style/Type: Technical Report / Team Description Paper (Experimental and System Design)

Cross References: The robotic system software design of the First order robot deviates from the standard performance first platforms like those used in Smit [2011] by prioritizing agile development though a python based system rather than a C++ system. This design aligns with the MICROMVP in Yu et al. [2017], however the navigation logic used by The first Order team is much more sophisticated. They chose to implement the Moving window approach and bisector algorithms to hand the dynamic movement required in SSL.

Summary: This paper addresses the engineering challenges of building a highly agile, autonomous robot for the RoboCup SSL competition, specifically overcoming the First Order team's past mechanical and software roadblocks. The core contribution is a complete mechatronic and software redesign. The new hardware features a compact 4-wheel omnidirectional base powered by DM3519 BLDC motors, a custom spring-damped dribbler system with closed-loop encoder feedback, and a high-voltage kicker utilizing a custom-wound solenoid and a rapidly charging capacitor array. The control stack was transitioned to Python for rapid development, heavily utilizing simulators like grSim and rSoccer for testing. The software architecture employs a thread-safe queue system, a centralized "Game Object" for state estimation, and a modified Dynamic Window Approach (DWA) paired with a bisector algorithm for highly reactive path planning. Preliminary testing demonstrated functional ball tracking while in motion and a kicking speed of roughly 3m/s. However, the system faces limitations, including an unfinished high-power kicker control board, an initial lack of GPIO pins requiring an extension board, and the need for further tuning to mitigate skidding and tipping at high speeds. The key takeaway is that leveraging a highly modular approach—both in hardware components and software roles/skills—provides a robust, expandable foundation for complex, high-speed robotic coordination.

Yu et al. [2017]
Jingjin Yu, Shuai D Han, Wei N Tang, and Daniela Rus. A portable, 3d-printing enabled multi-vehicle platform for robotics research and education. In *2017 IEEE International Conference on Robotics and Automation (ICRA)*, pages 1475–1480. IEEE, 2017

Aim: To create a portable and affordable micro scale robot that is suitable for research and teaching purposes

Style/Type: Conference Proceeding, Experimental

Cross References: This paper describes a way to build a micro scale robot that is programmable, referred to as MICROMVP. This design allows for quick mvp production that can be used to verify software implementation.However the dimensions of the MICROMVP are to small when compared to the allowed dimension fo the SSL.unlike Liang et al. [2025] and Smit [2011] which used omnidirectional wheel base the MICROMVP uses a 2 wheel-differential system for the wheelbase which aligns with the project scope

Summary: this paper introduces the MICROMVP, a mico-scale multi-vehicle platform, that is affordable, open source and easy to produce robotic test bed for research or educational purposes. This robot is built using 3D printing and off the shelf components making it's cost minimal. The MICROMVP is 8 X 5 X 6cm and weighs less than 100g. The robot uses a Spark Fio v3 board, XBee Series 1 for wireless communication and a dual motors to drive the wheels. The tracking systems is set up using a USB camera and chilitags fiducial markers to estimate states in SE for up to 50 robots simultaneously . The software is built using python for cross-platform compatibility and uses a subscriber-publisher model powered by ZeroMQ to handle high frequency feedback control loops (100Hz). The capabilities of the robot are demonstrated by its performance in several applications. The platform successfully utilized the pure pursuit algorithm for waypoint tracking and validated complex centralized (MINMAXDIST) and distributed (Reciprocal Velocity Obstacles) multi-robot path planning algorithms. The key takeaway is that the MICROMVP addresses the limitations of current solutions like the e-puck or Kilobots, these models are expense require more effort to assemble than the MICROMVP.Thus the MICROMVP is a robust too for validation complex multi-agent algorithms in constraint spaces.

Smit [2011]
Albert Smit. *Development of a robot for RoboCup Small Size League, utilizing a distributed control architecture for a multi-robot system development platform.* PhD thesis, Stellenbosch: Stellenbosch University, 2011

Aim: To address the lack of an existing research platform for fully autonomous, distributed robotic behaviour at Stellenbosch University, this thesis aims to design and manufacture the first prototype of an omni-directional soccer robot (Aim) for the domain of the RoboCup Small Size League (SSL) and broader Multi-Robot Systems (MRS) research. The approach involves bypassing the traditional centralized control architecture by outfitting the robot with a Single Board Computer (SBC) running Linux, custom motor controllers, and utilizing "Player" middleware to achieve a distributed control architecture

Style/Type: Thesis / Book, Experimental

Cross References: this thesis serves as a foundation, as it contains a lot of the initial data and steps taken to build a viable model. The produces model did not meet competition standards due to its dimension, additionally it was built based using an omni-wheel design unlike the Yu et al. [2017] which was designed to have a differential wheel system. from the software side the system uses the open source "Player" middle layer unlike the robot built in Joseph [2018] which uses the ROS system as a middle layer.

Summary: The thesis aims to develop a system that can replace the standard centralized architecture of traditional SSL robots which limits research in fully autonomous, decentralized multi-robot systems. A full testbed was developed from scratch, including a custom chassis, electronic stack powered by a Roboard RB 100 SBC and a C++ software API integrated into the "Player" middleware to handle the communication. The test bed was aimed to be tested in the RoboCup Small Size League and Muli-Robot Systems domains. The physical prototype was produced and passed three bench mark tests, a PI speed control test, a directional accuracy test for omni-wheels and a

static communication test that verified that the robots could communicate over the distributed middleware network. The robot however had its limitations, it was 4mm too large (184mm) to meet SSL competition guide lines (180mm).The open-loop position tracking had inaccuracies between 0.64 - 21 degrees due to wheel slippage. The scope of the project did not include the development of a dribbler or kicker mechanism, as well as a vision system, which are necessary features for a robot to partake in the SSL. With all the limitations in mind, this project successfully integrated mechanical , electronic and software systems into a highly modular and extendable distributed robotic platform that will serve as a testbed for future research.

Yao et al. [2015]
Weijia Yao, Wei Dai, Junhao Xiao, Huimin Lu, and Zhiqiang Zheng. A simulation system based on ros and gazebo for robocup middle size league. In *2015 IEEE international conference on robotics and biomimetics (ROBIO)*, pages 54–59. IEEE, 2015

Aim: To address the high cost and difficulty of testing multi-robot collaboration algorithms on physical robots, this paper aims to design and implement a virtual robot soccer simulation system for the domain of the RoboCup Middle Size League (MSL). The approach involves leveraging the open-source ROS and Gazebo platforms to build simplified 3D virtual models, utilizing a custom nubot gazebo model plugin to abstract hardware motions and bridge real-world robot code into the simulation

Style/Type: Conference Proceedings, Experimental

Cross References: The system is similar to Joseph [2018] in which it uses ROS to act as a middle ware to translate the directions into input that the simulation can interoperate and act on. Unlike Smit [2011] and Liang et al. [2025] which heavily rely on physical hardware to test their software this paper demonstration a way to test software algorithms through simulation allowing for simultaneous development on software and hardware.

Summary: The limitations of testing multi-robot collaboration algorithms on real robots is expensive and difficult. Existing simulators are proprietary or not widely used in MSL. This paper illustrates how a complete 3D simulation system was developed using ROS and Gazebo. A nu-bot gazebo plugin was created, this plugin mathematically simulates basic motions (dribbling , kicking and omni-directional motion) to completely replace low-level hardware controllers. The networking issue was solved by creating a bridge node that allows multiple simulated robot to communicate with a central "Coach" program. The simulated system successfully integrated real NuBot robot code into the virtual models. The system passed single-robot motion tests such as dribbling, chasing and kicking a ball. It also passed multi-robot test like communicating with "Coach" program and playing a match with rival robot models.To achieve acquit performance the the simulation uses simplistic physical mechanisms. It does not model wheels, dribblers or cameras, it represents the robot as a simple 31kg cylinder with out joints.Currently there is not omni-directional wheel modeling either. Despite these limitations the system is a cost-effective virtual testbed and by injecting Gaussian noise into the system for realism, the system can be used for debugging multi-robot collaboration algorithms before physical deployment

Chavan and Minase [2015]
AV Chavan and JL Minase. Design of a differential drive mobile robot platform for use in constrained environments. *International Journal of Innovations in Engineering Research and Technology*, 2(6):1–10, 2015

Aim: To address the challenge of robot maneuverability in highly congested spaces and the high cost of purchasing dedicated research platforms, this paper aims to design and develop a multitasking, differential-drive mobile robotic platform for use in constrained environments like hospital wards, offices, and warehouses. The approach involves designing a rectangular chassis in CATIA V5R19 equipped with two independent front driving DC motors and two rear caster wheels, and controlling the system using an onboard Atom PC and custom motor controllers

Style/Type: Journal Article, Experimental / System Design

Cross References: this paper, like Yu et al. [2017], is designed to produce a test bed for researcher to use for rapid development. Both designs papers also utilize a 2 wheel differential drive train to prover the motion of the robots. Unlike the robots designed in Smit [2011] and Liang et al. [2025] which use 4 wheel omni-wheel design to achieve holonomic movement, the robot uses kinematic equations to allow the robot to turn in place, despite it's front wheel design, to simulate the ability to move in in any direction from a single point.

Summary: The paper aims to develop a solution to the high cost of robot mobility testing in busy working environments such as hospitals and warehouses. Commercial solutions are expensive and makes testing algorithms in these environments challenging. A solution was designed using mathematical modeling and then physically built. The robot is designed using a differential drive train which powers the front wheels, the back wheels are caster wheels fitted with springs for shock absorption. The robot has a rectangular form factor and the wheels are driven using 2 Front-mounted Pittman DC motors with encoders and a control stack powered by an Atom PC with Cheetah-CB motor controllers. The robot was used to test the Kinematic equations, which proved to be correct as the robot successfully performed straight -line motion, rotation in place and executed dynamic turns. The robot is limited by the differential drive design as it lacks holonomic movement. This can be solved by switching to omni-wheels. Additionally the rectangular shape of the body changes the robots collisions footprint during turns, which is not an issue for robots with a circular design. The robots design balances the contains of weight, cost and bulkiness. The differential drive design provides a great open source testbed for researchers to easily implement and test advanced algorithms like path planning or dynamic steering and image recognition in tight spaces.

Joseph [2018]
Lentin Joseph. *Robot operating system (ros) for absolute beginners*, volume 2018. Springer, 2018

Aim: This book addresses the complexity of robotics software development by providing a step-by-step guide to programming custom ROS nodes. Within an educational framework, it details the use of standard ROS libraries for creating workspaces and publisher/subscriber nodes. Finally, it demonstrates hardware integration by bridging ROS with microcontrollers such as the Arduino and Raspberry P

Style/Type: Book Chapter, Technical Guide / Tutorial.

Cross References: this book serves as a great starting point for building a robot ROS as its middle ware. This contrasts Smit [2011] which mentions ROS as a middle ware but chooses to implement the "Player" middleware instead. This book also compliments Yao et al. [2015] as they both use ROS for simulating robots and their behaviour, however unlike Yao et al. [2015] the book also goes into how to build physical robots with ROS as a middleware. Additionally Unlike Liang et al. [2025] which uses custom circuit board to implement their software the book demonstrates how to use ROS with off the shelf circuit boards like an Arduino board, this helps keep the build cost-effective.

Summary: The book teaches beginner how to translates high level robotic concepts into working code by introducing concepts like hardware abstraction, multi-process communication and package management. Through comprehensive tutorials on C++ and python one is taught how to build ROS applications. It covers setting up a build system,structuring files and writing nodes that publish and subscribe to data streams. By one progress through the book, they will pass basic messages through nodes, move a 2D and 3D virtual robot though kinematic commands, learn to read global position feedback and bridge ROS to a physical board using different packages. Due to the book being targeted at beginners is does not cover more complex topics such as muli-robot systems and computer vision algorithms that would be required for Robocup's SSL. However the book provides a foundation to develop the software architecture and

hardware integration techniques that are required to build the ROS compatible robot using off the shelf micro-controllers and other low-cost parts.

References

AV Chavan and JL Minase. Design of a differential drive mobile robot platform for use in constrained environments. *International Journal of Innovations in Engineering Research and Technology*, 2(6):1–10, 2015.

Lentin Joseph. *Robot operating system (ros) for absolute beginners*, volume 2018. Springer, 2018.

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Weijia Yao, Wei Dai, Junhao Xiao, Huimin Lu, and Zhiqiang Zheng. A simulation system based on ros and gazebo for robocup middle size league. In *2015 IEEE international conference on robotics and biomimetics (ROBIO)*, pages 54–59. IEEE, 2015.

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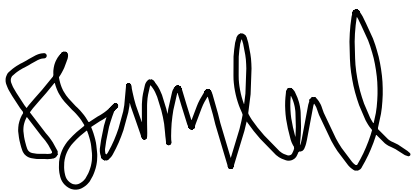
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